



When we draw a graph on a piece of paper, we naturally try to do this as transparently as possible. One obvious way to limit the mess created by all the lines is to avoid intersections. For example, we may ask if we can draw the graph in such a way that no two edges meet in a point other than a common end.

Graphs drawn in this way are called *plane graphs*; abstract graphs that *can* be drawn in this way are called *planar*. Due to their natural appeal, plane graphs have been studied right from the outset of graph theory; the most prominent example is the four colour problem for maps, which we shall meet in Chapter 5.

The seemingly unrelated question of how we can characterize planar graphs combinatorially, and conversely of which role they play elsewhere in graph theory, is not only equally interesting, but has been found to be quite unexpectedly relevant. Indeed, we shall see in Chapter 12 that the planar graphs – in fact, the graphs embeddable in any given surface – play a central role in one of the deepest theorems of all of graph theory: the structure theorem that describes all graphs without any fixed minor. (Note that there is nothing topological in the definition of that class.) Thus, planar graphs are not just for colouring puzzles: they are important, and they deserve rigorous study.

In this chapter we study both plane and planar graphs, as well as the relationship between the two: the question of how an abstract planar graph might be drawn in fundamentally different ways. After collecting together in Section 4.1 the few basic topological facts that will enable us later to prove all results rigorously without too much technical ado, we begin in Section 4.2 by studying the structural properties of plane graphs. In Section 4.3, we investigate how two drawings of the same graph can differ. The main result of that section is that 3-connected planar graphs have essentially only one drawing, in some very strong and natural topological sense. The next two sections are devoted to the

proofs of all the classical planarity criteria, conditions telling us when an abstract graph is planar. We complete the chapter with a section on *plane duality*, a notion with fascinating links to algebraic, colouring, and flow properties of graphs (Chapters 1.9 and 6.5).

The traditional notion of a graph drawing is that its vertices are represented by points in the Euclidean plane, its edges are represented by curves between these points, and different curves meet only in common endpoints. To avoid unnecessary topological complication, however, we shall only consider curves that are piecewise linear; it is not difficult to show that any drawing can be straightened out in this way, so the two notions come to the same thing.

4.1 Topological prerequisites

In this section we briefly review some basic topological definitions and facts needed later. All these facts have (by now) easy and well-known proofs; see the notes for sources. Since those proofs contain no graph theory, we do not repeat them here: indeed our aim is to collect precisely those topological facts that we need but do *not* want to prove. Later, all proofs will follow strictly from the definitions and facts stated here (and be guided by but not rely on geometric intuition), so the material presented now will help to keep elementary topological arguments in those proofs to a minimum.

A *straight line segment* in the Euclidean plane is a subset of $\mathbb{R}^2$ that has the form $\{p + \lambda(q - p) \mid 0 \leq \lambda \leq 1\}$ for distinct points $p, q \in \mathbb{R}^2$. A *polygon* is a subset of $\mathbb{R}^2$ which is the union of finitely many straight line segments and is homeomorphic to the unit circle S^1 , the set of points in $\mathbb{R}^2$ at distance 1 from the origin. Here, as later, any subset of a topological space is assumed to carry the subspace topology. A *polygonal arc* is a subset of $\mathbb{R}^2$ which is the union of finitely many straight line segments and is homeomorphic to the closed unit interval $[0, 1]$. The images of 0 and of 1 under such a homeomorphism are the *endpoints* of this polygonal arc, which *links* them and runs *between* them. Instead of ‘polygonal arc’ we shall simply say *arc* in this chapter. If P is an arc between x and y , we denote the point set $P \setminus \{x, y\}$, the *interior* of P , by $\mathring{P}$. As continuous images of $[0, 1]$, arcs, and finite unions of arcs, are compact, and hence closed in $\mathbb{R}^2$. Their complements in $\mathbb{R}^2$, therefore, are open.

Let $O \subseteq \mathbb{R}^2$ be any open set. Being linked by an arc in O defines an equivalence relation on O . The corresponding equivalence classes are again open; they are the *regions* of O . A closed set $X \subseteq \mathbb{R}^2$ is said to *separate* a region O' of O if $O' \setminus X$ has more than one region. The *frontier* of a set $X \subseteq \mathbb{R}^2$ is the set Y of all points $y \in \mathbb{R}^2$ such that every neighbourhood of y meets both X and $\mathbb{R}^2 \setminus X$. Note that if X is open then its frontier lies in $\mathbb{R}^2 \setminus X$.

polygon

arc

$\mathring{P}$, $\mathring{e}$

region

separate

frontier

The frontier of a region O of $\mathbb{R}^2 \setminus X$, where X is a finite union of points and arcs, has two important properties. The first is accessibility: if $x \in X$ lies on the frontier of O , then x can be linked to some point in O by a straight line segment whose interior lies wholly inside O . As a consequence, any two points on the frontier of O can be linked by an arc whose interior lies in O (why?). The second notable property of the frontier of O is that it separates O from the rest of $\mathbb{R}^2$. Indeed, if $\varphi: [0, 1] \rightarrow P \subseteq \mathbb{R}^2$ is continuous, with $\varphi(0) \in O$ and $\varphi(1) \notin O$, then P meets the frontier of O at least in the point $\varphi(y)$ for $y := \inf \{x \mid \varphi(x) \notin O\}$, the *first point* of P in $\mathbb{R}^2 \setminus O$.

Theorem 4.1.1. (Jordan Curve Theorem for Polygons)

For every polygon $P \subseteq \mathbb{R}^2$, the set $\mathbb{R}^2 \setminus P$ has exactly two regions. Each of these has the entire polygon P as its frontier.

With the help of Theorem 4.1.1, it is not difficult to prove the following lemma.

Lemma 4.1.2. Let P_1, P_2, P_3 be three arcs, between the same two end-points but otherwise disjoint.

- (i) $\mathbb{R}^2 \setminus (P_1 \cup P_2 \cup P_3)$ has exactly three regions, with frontiers $P_1 \cup P_2$, $P_2 \cup P_3$ and $P_1 \cup P_3$.
- (ii) If P is an arc between a point in $\overset{\circ}{P}_1$ and a point in $\overset{\circ}{P}_3$ whose interior lies in the region of $\mathbb{R}^2 \setminus (P_1 \cup P_3)$ that contains $\overset{\circ}{P}_2$, then $\overset{\circ}{P} \cap \overset{\circ}{P}_2 \neq \emptyset$.

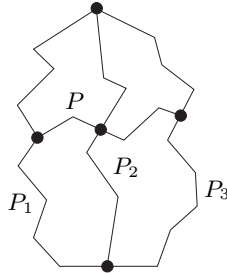


Fig. 4.1.1. The arcs in Lemma 4.1.2 (ii)

Our next lemma complements the Jordan curve theorem by saying that an arc does *not* separate the plane. For easier application later, we phrase this a little more generally:

Lemma 4.1.3. Let $X_1, X_2 \subseteq \mathbb{R}^2$ be disjoint sets, each the union of finitely many points and arcs, and let P be an arc between a point in X_1 and one in X_2 whose interior lies in a region O of $\mathbb{R}^2 \setminus (X_1 \cup X_2)$. Then $O \setminus P$ is a region of $\mathbb{R}^2 \setminus (X_1 \cup P \cup X_2)$.

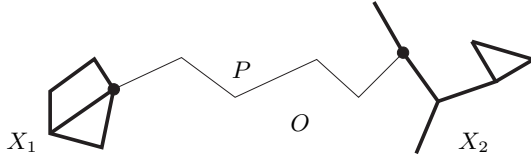


Fig. 4.1.2. P does not separate the region O of $\mathbb{R}^2 \setminus (X_1 \cup X_2)$

It remains to introduce a few terms and facts that will be used only once, when we consider notions of equivalence for graph drawings in Chapter 4.3.

S^n As usual, we denote by S^n the n -dimensional sphere, the set of points in $\mathbb{R}^{n+1}$ at distance 1 from the origin. The 2-sphere minus its ‘north pole’ $(0, 0, 1)$ is homeomorphic to the plane; let us choose a fixed such homeomorphism $\pi: S^2 \setminus \{(0, 0, 1)\} \rightarrow \mathbb{R}^2$ (for example, stereographic projection). If $P \subseteq \mathbb{R}^2$ is a polygon and O is the bounded region of $\mathbb{R}^2 \setminus P$, let us call $C := \pi^{-1}(P)$ a *circle on S^2* , and the sets $\pi^{-1}(O)$ and $S^2 \setminus \pi^{-1}(P \cup O)$ the *regions* of $S^2 \setminus C$.

Our last tool is the theorem of Jordan and Schoenflies, again adapted slightly for our purposes:

[4.3.1] **Theorem 4.1.4.** *Let $\varphi: C_1 \rightarrow C_2$ be a homeomorphism between two circles on S^2 , let O_1 be a region of $S^2 \setminus C_1$, and let O_2 be a region of $S^2 \setminus C_2$. Then φ can be extended to a homeomorphism $C_1 \cup O_1 \rightarrow C_2 \cup O_2$.*

4.2 Plane graphs

plane graph A *plane graph* is a pair (V, E) of finite sets with the following properties (the elements of V are again called *vertices*, those of E *edges*):

- (i) $V \subseteq \mathbb{R}^2$;
- (ii) every edge is an arc between two vertices;
- (iii) different edges have different sets of endpoints;
- (iv) the interior of an edge contains no vertex and no point of any other edge.

A plane graph (V, E) defines a graph G on V in a natural way. As long as no confusion can arise, we shall use the name G of this abstract graph also for the plane graph (V, E) , or for the point set $V \cup \bigcup E$; similar notational conventions will be used for abstract versus plane edges, for subgraphs, and so on.¹

¹ However, we shall continue to use $\setminus$ for differences of point sets and $-$ for graph differences – which may help a little to keep the two apart.